# Cameron DeBough

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## EDUCATION

**University of Arizona**, B.S. Computer Science & Artificial Intelligence, Minor in Mathematics **Expected May 2029**
GPA: 3.87 · 67 Credits Completed
**Honors:** Dean's List with Distinction (Fall 2025) · Dean's List (Spring 2026) · Undergraduate Research Travel Grant, $1,000 (Fall 2026)
**Coursework:** Software Development, Discrete Mathematics for CS I-II, Analyzing Data, Web Programming, Calculus I-II, Linear Algebra

## EXPERIENCE

**Software Engineering Intern**, SkillfullyAware / SAAQ **October 2025 - Present**
- Own the production backend of an AI-powered leadership assessment platform used by executive peer organizations (EO, YPO): Python and FastAPI services on Supabase (PostgreSQL), live in production
- Built a streaming LLM analysis pipeline with multi-stage scoring and two-pass generation that eliminated timeouts and truncated output, plus automated DOCX report generation from the analysis results
- Implementing authentication, Stripe payments, and persistent user storage for MVP launch

**Founder & Engineer**, DeBough Systems (deboughsystems.com) **August 2026 - Present**
- Built and launched a cybersecurity service that helps small businesses with no IT staff find and fix the weak spots attackers target most, with every issue explained in plain English
- Wrote a read-only checkup program that scans each company computer in about 40 seconds (updates, antivirus, firewall, disk encryption) and turns the results into a ranked fix list, a weekly email report, and a printable report for cyber insurers
- Built a free domain scan that checks whether attackers could fake a company's email or impersonate it with look-alike websites; Next.js, TypeScript, Supabase/Postgres, and Stripe, with rate limiting and ownership checks to prevent misuse

**Research Software Engineer**, NewFOS Center, University of Arizona **March 2026 - Present**
*New Frontiers of Sound, an NSF Science & Technology Center*
- Built Python automation pipelines for nonlinear acoustics experiments (hysteresis sweeps, harmonic SNR analysis, transfer-function extraction) used to model signal distortion in acoustic transducers
- Designed and validated an electrical impedance measurement rig with a full de-embedding pipeline that isolated a 42% cable-parasitic error in a previously reported measurement
- Developed MATLAB signal-analysis pipelines (SVD, dynamic mode decomposition) that produced publication-quality figures for experiments on a three-rod coupled acoustic waveguide
- Automated Tektronix oscilloscopes and function generators via VISA/SCPI for synchronized multi-channel data acquisition

**Undergraduate Research Assistant**, VSI Lab, University of Arizona ECE **September 2025 - Present**
*Embodied AI & Accessibility · Advisor: Prof. Eungjoo Lee*
- Led a real-robot study comparing online Gaussian Splatting SLAM to offline COLMAP/GLOMAP mapping on a TurtleBot 4; corrected an 11-13% depth camera bias using lidar, cutting tracking error from 7.4 cm to 1.2 cm
- Building NVIDIA Isaac Sim environments for robot inspection with ROS 2 control of LeRobot SO-101 arms
- Building computer-vision pipelines for Meta Aria 2 smart glasses (hand tracking, gaze data, real-time on-device ML inference) for ASL accessibility and skill-assessment applications

## TECHNICAL PROJECTS

**APEX QUANT**, Algorithmic Futures Trading System
- Architected event-driven backtesting and paper-trading engine for micro futures (MES/MNQ/MGC) with modular strategy framework, risk management controls, and a broker-integration layer targeting Interactive Brokers (ib_insync)

**Thermal-Tracking Turret**, Embedded Systems / Robotics
- Built an autonomous heat-tracking pan/tilt turret on Raspberry Pi: AMG8833 8×8 thermal array over I2C, PCA9685 hardware PWM servo control, and a closed-loop feedback algorithm that centers a moving heat source in real time, deployed as a headless systemd service

**Synapse Finance**, Full-Stack Financial SaaS Platform
- Shipped production app with 50+ active users: Flask, React, PostgreSQL, JWT authentication, RESTful APIs, and CI/CD

## PRESENTATIONS

DeBough, C., Koh, W., Lee, E. "Online Gaussian Splatting SLAM on a TurtleBot4: Real-Robot Evaluation Against Offline Structure-from-Motion." Poster, *Bay Area Robotics Symposium (BARS)*, UC Berkeley, October 2026

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, JavaScript/TypeScript, SQL, MATLAB, Bash, HTML/CSS
**Frameworks:** FastAPI, Flask, React, Next.js, Node.js, Tailwind CSS
**Embedded & Hardware:** Raspberry Pi, I2C sensors, PWM servo control, VISA/SCPI instrument automation, ROS 2
**AI & Data:** Agentic AI (tool calling, multi-step agents, RAG), LLM integration, computer vision, NumPy, SciPy, Pandas, signal processing
**Tools & Databases:** Git, Docker, Linux, PostgreSQL, Supabase, Stripe, CI/CD

## LEADERSHIP & ACTIVITIES

**Coke Fellow**, The Coca-Cola Company, Introships Career Discovery Program **May - July 2026**
- Selected as 1 of 50 fellows nationwide for a coached cohort program with Coca-Cola executives

**Founder & Owner**, Durian Car Detailing **May 2023 - Present**
- Built and run a car detailing business from scratch with 200+ jobs completed, handling pricing, marketing, and customer service

**University of Arizona AI Club** · MLH Global Hack Week · 13+ GitHub projects